Consider a Neural network:

Input: single scalar x

Layer 1: $h = \tanh(w_1 x + b_1)$

Layer 2: $o = w_2 h + b_2$

Loss: $L = (o - y)^2$ where y is the true value

$$\frac{\partial L}{\partial b_2} = \frac{\partial (w_2 h - b_2 - y)^2}{\partial b_2} = 2(w_2 h + b_2 - y)(+1) = +2(w_2 h - b_2 - y)$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial (w_2 h - b_2 - y)^2}{\partial w_2} = 2(w_2 h - b_2 - y)(h) = 2h(w_2 h - b_2 - y)$$

$$\frac{\partial L}{\partial o} = \frac{\partial (o - y)^2}{\partial o} = 2(o - y); \quad \frac{\partial L}{\partial h} = \frac{\partial L}{\partial o} * \frac{\partial o}{\partial h} = 2(o-y) * \frac{\partial (w_2 h + b_2)}{\partial h}$$

$$\Rightarrow 2(o-y) * w_2$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial h}{\partial w_1} * \frac{\partial L}{\partial h} \Rightarrow \frac{\partial (\tanh(w_1 x + b_1))}{\partial w_1} * \frac{\partial L}{\partial h}$$

$$\Rightarrow \left(1 - (\tanh(w_1 x + b_1))^2\right) * x \Rightarrow \left(1 - (\tanh(w_1 x + b_1))^2\right) * x * 2(o-y) * w_2$$

Similarly,

$$\frac{\partial L}{\partial b_1} = \frac{\partial h}{\partial b_1} * \frac{\partial L}{\partial h} = \left(1 - (\tanh(w_1 x + b_1))^2\right) * 2(o-y) * w_2$$